

Chanelle LeGrand

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PROFILE

A full-stack software engineer with hands-on experience integrating large language models (LLMs) into production applications and deploying secure cloud infrastructure on AWS. Comfortable across the stack — from prompt engineering and LLM API workflows to infrastructure as code — with a focus on translating ambiguous problems into reliable, scalable systems.

EDUCATION

Howard University, Washington, DC

Aug 2020 – Dec 2025

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Cloud Computing

TECHNICAL SKILLS

AI & LLM Integration: Prompt Engineering, Large Language Models (LLMs), LLM API Integration, Groq, Llama 3.3, Generative AI, Context Injection, Structured Output Parsing, AI-Powered Workflows

Cloud & Infrastructure: AWS (EC2, RDS, ALB, Route 53, VPC), Terraform, Docker, Linux, Bash/Shell Scripting, Railway, Vercel

Backend & APIs: Python, FastAPI, Node.js, Express.js, REST APIs, SQL, Async I/O

Frontend: React, TypeScript, Vite, Tailwind CSS, Vanilla JavaScript

Languages: Python, JavaScript, TypeScript, Java, C, C++

DevOps & Automation: GitHub Actions, Docker Compose, CI/CD

EXPERIENCE

Python Tutor, Howard University

Fall 2024

- Supported 20+ students in Python I–III with debugging and core programming concepts
- Reviewed assignments and improved students' problem-solving approach
- Reinforced clean code practices and modular design

PROJECTS

Pulse | Full-Stack AI Pipeline Intelligence Platform

Apr 2026 – May 2026

Lead Developer | Full-stack B2B sales pipeline dashboard with AI-powered re-engagement email drafting

Tech Stack: React, TypeScript, Vite, Tailwind CSS, Python, FastAPI, Groq API (Llama 3.3 70B), Railway, Vercel

- Built a B2B sales pipeline risk engine that scores CRM deals on a 0–100 scale across 5 weighted rules — customer silence, stage stagnation, stale next steps, single threading, and slipped close dates — auto-bucketing each into a 3-tier red/yellow/green system surfaced through 4 FastAPI REST endpoints
- Integrated the Groq API (Llama 3.3 70B) to generate AI-powered re-engagement email drafts on demand, dynamically injecting deal stage, dollar amount, top 3 risk drivers, and recent email thread history into structured prompts for contextual grounding
- Engineered prompt templates with explicit role assignment, 3–4 sentence length constraints, tone guidance, and a structured Subject/Body output schema, plus a downstream output parser with fallbacks that guarantee well-formed responses across model deviations
- Developed a React + TypeScript dashboard (Vite, Tailwind) over a 200-deal demo dataset featuring a sortable risk-scored deal table, pipeline-at-risk KPI cards, a rep leaderboard, and a one-click AI draft flow rendered into the deal detail panel

SafeRouteDC | Full-Stack Cloud Application

Nov 2025 – Feb 2026

Lead Developer & Cloud Architect | Pedestrian safety app deployed on AWS generating real-time route risk predictions for Washington, DC

Tech Stack: Node.js, Express.js, REST APIs, AWS (EC2, RDS, ALB, Route 53, VPC), Docker, Linux

- Designed and deployed a production full-stack application on a private AWS EC2 instance with a Node.js/Express.js backend, eliminating all public ingress to the host; ~78% of Howard University student users reported traveling more safely after adopting the tool
- Architected a secure 3-tier VPC with a public-facing ALB and private EC2/RDS subnets, enforcing least-privilege Security Groups that restrict inbound traffic to ALB-only, and routed administrative SSH through a VPC endpoint instead of exposing port 22 publicly
- Designed a custom safety-scoring algorithm fusing a 1–5 crime severity scale with time-decay weighting